

**BUSINESS ANALYSIS  
PORTFOLIO CASE STUDY**

# **BNPL Platform Performance & Transaction Approval Enhancement**

End-to-End Business Analysis Case Study

A simulated FinTech case study investigating declining transaction approval performance, increasing processing time, and identity-verification reliability.

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Business Analysis | Performance Analytics | Process Improvement

Tools: Excel • Power BI • SQL • Jira • Draw.io • DAX

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# 1. Executive Summary

This case study demonstrates an end-to-end Business Analysis approach for a simulated Buy Now, Pay Later (BNPL) platform. The analysis focused on a three-month deterioration in transaction performance, with higher customer and merchant complaints, lower approval performance, and slower processing.

| Metric                  | Result   |
|-------------------------|----------|
| Transactions Analyzed   | 6,000    |
| Overall Approval Rate   | 71.7%    |
| Average Processing Time | 5.11 sec |
| Technical Errors        | 396      |

## Project objective

Identify the causes of deteriorating BNPL transaction performance, define measurable business and system requirements, redesign the affected process, and establish an Agile delivery, testing, monitoring, and solution-evaluation approach.

## My role in the case study

- Defined the business problem, objectives, scope, stakeholders, and success measures.
- Validated data quality and analyzed transaction KPIs using Excel, Power BI, DAX, and basic SQL.
- Performed root-cause analysis and stakeholder elicitation across Operations, Product, Risk, Engineering, QA, and Data.
- Produced requirements, AS-IS / TO-BE process designs, gap analysis, BRD content, Agile backlog items, UAT scenarios, and solution-evaluation criteria.

**Note:** This is an independent portfolio case study using simulated data. It does not represent production work performed for a real BNPL provider.

## 2. Business Problem, Objectives & Scope

### Problem statement

Over the past three months, the BNPL platform experienced increased customer and merchant complaints related to declined and delayed transactions. Management reported deterioration in approval performance and transaction processing time, with negative effects on customer experience, merchant satisfaction, conversion, and GMV growth. The root causes were initially unknown.

| Business Objective                         | Success Measure                                       |
|--------------------------------------------|-------------------------------------------------------|
| Improve transaction approval performance   | Approval Rate ≥ 85%                                   |
| Improve transaction processing performance | Average Processing Time ≤ 5 seconds                   |
| Maintain required risk controls            | No removal of required identity-verification controls |
| Improve technical reliability              | Eliminate unintended duplicate verification requests  |

### Scope

| In Scope                                                                                                                          | Out of Scope                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Approval workflow; identity-verification calls; exception handling; monitoring; change impact assessment; E2E regression testing. | UI redesign; merchant settlement; changes to core risk appetite or risk policy unless separately approved. |

### 3. Data Quality & Performance Analysis

A dataset of 6,000 simulated BNPL transactions covering March-August 2026 was reviewed before analysis. The raw data was preserved, a clean-data layer was created, and a Data Quality Log documented each correction.

| Issue                       | Validated Correction                              |
|-----------------------------|---------------------------------------------------|
| Missing Decline Reason      | Technical Error                                   |
| Negative Transaction Amount | -350 → 350                                        |
| Negative Approval Time      | -2.5 → 2.50 sec                                   |
| Duplicate transaction IDs   | Three extraction duplicates reassigned unique IDs |

#### Key findings

- Approval Rate declined from 76.4% in Mar-May to 66.9% in Jun-Aug: a 9.5 percentage-point deterioration.
- Technical-error declines increased from 123 in Mar-May to 273 in Jun-Aug: approximately +122%.
- Mobile App technical errors increased from 58 to 203: approximately +250%.
- Mobile App Approval Rate declined from 74.7% to 60.2%: a 14.5 percentage-point deterioration.
- Mobile App average processing time increased from approximately 3.81 sec to 6.99 sec: approximately +83.5%.

# 4. Power BI Performance Dashboard

The dashboard consolidates the main business KPIs and root-cause indicators into a single executive view. It supports filtering by period, channel, and merchant type, and compares actual performance against the defined approval-rate and processing-time targets.

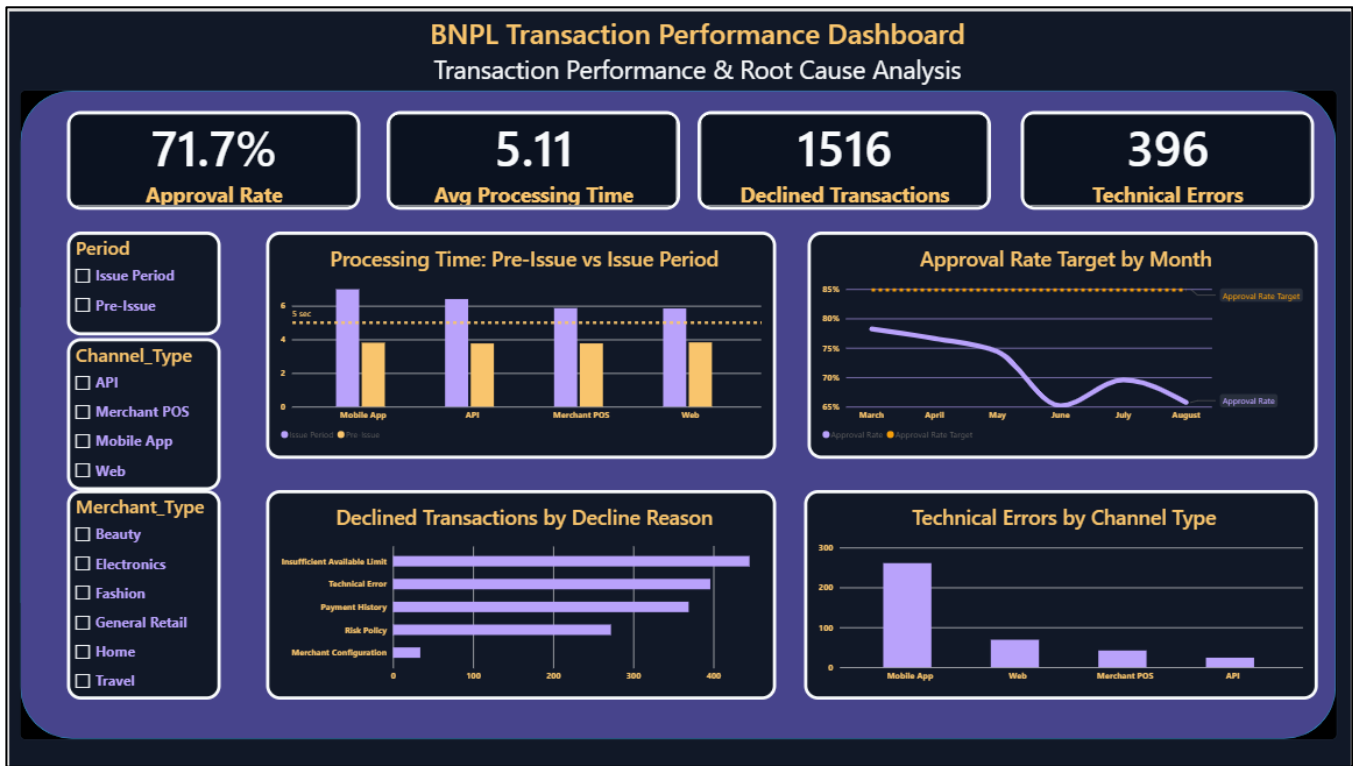


Figure 1. Updated BNPL Transaction Performance Dashboard

## Dashboard story

The visual flow moves from overall KPI performance to trend and period comparison, then to diagnostic views of decline reasons and technical errors by channel. The dashboard is intended to support investigation and post-implementation monitoring rather than to imply that all declines are technical failures.

# 5. Root Cause Analysis

Stakeholder elicitation identified a major change around the start of the issue period: additional real-time risk checks were introduced in the transaction decisioning service. Engineering evidence showed that the external identity-verification integration also became materially slower, while some Mobile App flows could trigger verification through both an existing checkout path and the new risk workflow.

| Evidence                       | Observation                                                                                       |
|--------------------------------|---------------------------------------------------------------------------------------------------|
| Decisioning service            | Average processing component increased from ~1.4 sec to ~2.1 sec after new checks                 |
| External identity verification | Average response increased from ~0.8 sec to ~2.6 sec                                              |
| Mobile App flow                | Existing checkout verification and new risk-workflow verification could both occur                |
| Testing / release              | Channel-specific dependency was not fully identified in impact assessment or E2E regression scope |

**Root Cause:** An incomplete change impact assessment failed to identify the existing Mobile App identity-verification dependency. The new risk workflow therefore introduced an additional verification call in some flows, while E2E regression testing did not cover the affected interaction.

## 6. Process Analysis: AS-IS, Gaps & TO-BE

| Gap    | AS-IS                                                                         | TO-BE                                                                                           |
|--------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| GAP-01 | Multiple verification requests may occur for the same Mobile App transaction. | Prevent unintended duplicate verification requests; authorized retry remains permitted.         |
| GAP-02 | No clearly defined timeout / unavailable-service handling.                    | Retry once; if unsuccessful, set transaction to Pending and inform customer.                    |
| GAP-03 | Insufficient monitoring of verification performance and failures.             | Monitor response time, timeouts, failures, and request behavior.                                |
| GAP-04 | Impact assessment and E2E regression missed channel dependency.               | Identify affected channels/integrations/dependencies and test affected flows before production. |

### TO-BE transaction logic

Customer selects BNPL → transaction/customer information is validated → risk criteria are evaluated → if identity verification is required, one legitimate verification request is initiated → if the service fails or times out, one authorized retry is permitted → if the retry also fails, the transaction becomes Pending and the customer is informed that processing is temporary → decisioning continues when verification succeeds.

## 7. Requirements & Traceability

| ID       | Type                 | Requirement / Criterion                                                                                                                            |
|----------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-01    | Business             | Approval Rate $\geq 85\%$ while average transaction processing time $\leq 5$ sec.                                                                  |
| BRULE-01 | Business Rule        | If a transaction meets defined risk criteria, identity verification is triggered.                                                                  |
| FR-01    | Functional           | Prevent unintended duplicate verification requests for the same transaction in one approval flow; authorized retry under FR-02 is not a duplicate. |
| FR-02    | Functional           | On timeout/unavailability, retry once; if retry fails, set Pending and inform customer.                                                            |
| FR-03    | Functional           | Monitor identity-verification response times, failures, and timeouts.                                                                              |
| NFR-01   | Performance          | Identity-verification service responds within 2 sec for at least 95% of requests.                                                                  |
| TR-01    | Transition           | Perform E2E regression testing across all affected channels before production.                                                                     |
| TR-02    | Transition / Process | Identify affected channels, integrations, and dependencies through change impact assessment before implementation.                                 |

Traceability was established conceptually from Business Need → Requirement → Agile Feature/User Story → Acceptance Criteria → Test Case → Result. Open items were retained rather than invented, including the final alert threshold/recipients for FR-03 and the final classification of TR-02.

## 8. Agile Delivery, Jira & Testing

### Epic: Improve Identity Verification Reliability and Performance

| Backlog Item                                     | Story Points | Sprint                 |
|--------------------------------------------------|--------------|------------------------|
| Prevent Duplicate Identity Verification Requests | 5            | Sprint 1               |
| Handle Identity Verification Service Failure     | 5            | Sprint 1               |
| Monitor Identity Verification Performance        | 3            | Backlog / later sprint |

The screenshot displays the Jira interface for the 'BNPL Platform Enhancement' project. The left sidebar shows the 'Backlog' view with a search bar and a filter. The main area shows the 'BNPL Sprint 1' (24 Sep – 8 Oct) with two work items: 'SCRUM-6 Prevent Duplicate Identity Verifica...' (5 points) and 'SCRUM-7 Handle Identity Verification Servic...' (5 points). Below the sprint is the 'Backlog' with one work item: 'SCRUM-8 Monitor Identity Verification Performan...' (3 points). The right sidebar shows a detailed view of the 'SCRUM-6' user story, titled 'Prevent Duplicate Identity Verification Requests'. The description includes a 'User Story' and 'Acceptance Criteria'.

**BNPL Platform Enhancement**

Summary | Backlog | Board | Calendar | Timeline | Docs | Forms

Search backlog | Filter

**BNPL Sprint 1** 24 Sep – 8 Oct (2 work items) | 0 | 10 | 0 | Complete sprint

Improve the reliability of the BNPL identity-verification flow by preventing duplicate verification requests and handling temporary verif...

SCRUM-6 Prevent Duplicate Identity Verifica... | Improve I... | To Do | 5

SCRUM-7 Handle Identity Verification Servic... | Improve I... | To Do | 5

+ Create

**Backlog** (1 work item) | 3 | 0 | 0 | Create sprint

SCRUM-8 Monitor Identity Verification Performan... | Improve I... | Idea | 3

+ Create

**Jira work item**

SCRUM-5 / SCRUM-6

**Prevent Duplicate Identity Verification Requests**

To Do | Improve Feature

**Description**

**User Story:**

As a BNPL Operations team member, I want the system to trigger no more than one identity-verification request for the same transaction during a single approval flow, so that duplicate verification requests are prevented and transaction processing remains efficient.

**Acceptance Criteria:**

**AC-01:** Given a transaction requires identity verification, when the transaction enters the identity-verification step, then the system shall send only one identity-verification request for

Figure 2. Jira Sprint Backlog and User Story - Translation of BNPL requirements into prioritized Agile work items with story points and acceptance criteria.

**Sprint Goal:** Improve the reliability of the BNPL identity-verification flow by preventing duplicate verification requests and handling temporary verification-service failures.

### Proposed UAT scenarios

- Verification required → one legitimate verification request is initiated.
- Another component attempts an unintended duplicate → duplicate is prevented.
- First request times out, retry succeeds → one authorized retry and the approval flow continues.
- First request and retry both fail → transaction is Pending.
- Pending due to verification failure → customer sees Temporarily Under Processing, not a credit/risk decline.
- Verification not required → no verification request.
- Monitoring captures response time, failures, and timeouts and supports evaluation against the performance target.

## 9. Solution Evaluation & Post-Implementation Plan

| KPI                               | Baseline / Current           | Success Criterion       | Monitoring         |
|-----------------------------------|------------------------------|-------------------------|--------------------|
| Approval Rate                     | 71.7% overall                | ≥85%                    | Weekly for 4 weeks |
| Avg Processing Time               | 5.11 sec overall             | ≤5 sec                  | Daily + weekly     |
| Unintended Duplicate Verification | Observed in affected flow    | 0                       | Daily              |
| Verification Response Time        | Degraded during issue period | ≤2 sec for ≥95%         | Daily              |
| Technical Errors                  | 396 in analyzed dataset      | Reduction from baseline | Weekly             |

### Evaluation principle

Implementation success and business-outcome success are evaluated separately. If technical targets are achieved but Approval Rate remains below 85%, the next BA step is to quantify the remaining gap, analyze residual decline reasons, separate technical from business/risk declines, and engage the appropriate stakeholders before proposing additional requirements.

## 10. Portfolio Summary

### What this case study demonstrates

- Business problem framing and measurable success criteria.
- Stakeholder analysis and evidence-based elicitation.
- Data quality validation, KPI analysis, and root-cause investigation.
- Business, stakeholder, rule, functional, non-functional, and transition requirements.
- AS-IS / TO-BE thinking, gap analysis, and BPMN process modeling.
- Agile translation into Epic, backlog items, acceptance criteria, sprint scope, and Jira workflow.
- Traceability, UAT planning, release readiness, and solution evaluation.
- Executive communication through a Power BI dashboard.